



Figure 2: Closed framework for morphological system description. A generative neural network synthesizes possible tomographies (CTs) of material composites or assemblies based on predefined manufacturing configurations. The morphological representation enables the visualization of various manufacturing influences. A surrogate model is based on the spatial representation and describes the physical properties. This finally enables an evaluation of the materials design

Finally, symbolic formula extraction algorithms and algorithms from the field of explainable AI are used to extract symbolic formulas. These learned formulas guide the design process, e.g. by helping to identify optimal materials designs for specific tasks or identifying promising candidates for targeted experiments.

### Morphological Representation

Due to a symbolic description, the physical materials behavior can be theoretically interpreted. However, the generated equations are often difficult to interpret in practice due to their complexity. Another way to better understand system relationships is to synthesize materials structures in the form of tomography. This approach is visualized in Figure 2.

The generative part uses CTs as training data. These are labeled with manufacturing configurations such as material compositions and process parameters. Under these high-dimensional conditions, a GAN or DDPM is trained to synthesize high-resolution 3D tomography data. In the operation phase, the respective influence on the morphological structure is analyzed by varying the conditions. Analogous to the unsupervised approach mentioned above, a surrogate model is then used to learn and represent the materials behavior. Finally, the design draft is evaluated.

### Conclusion

A comprehensive approach that integrates ML, CTs and experiments is presented. This solves the lack of transferable and interpretable materials laws for complex composite materials and assemblies. The surrogate design model provides an efficient way to overcome the traditional bottleneck in the design-test cycle.

The algorithmic representation ensures the interpretability of the neural network-based models. The extraction of explicit, formula-based models enhances understanding and facilitates meaningful contributions to materials design.

In addition, the morphological representation provides a visual understanding of the materials system through synthesis of 3D tomography data. This enables the analysis of

manufacturing influences on morphological structures and provides valuable insights into the material design process.

The proposed framework not only streamlines the materials design process but also empowers researchers with a powerful tool to explore and predict the behavior of complex material composites. By combining CTs synthesis and explainable AI, this modern approach opens opportunities for the rapid and informed design of materials with tailored properties.

### Acknowledgment

This work was supported by the project "KIBIDZ – Intelligente Brandgefahrenanalyse für Gebäude und Schutz der Rettungskräfte durch Künstliche Intelligenz und Digitale Brandgebäudezwillinge" by the Zentrum für Digitalisierungs- und Technologieforschung der Bundeswehr (dtec.bw).

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